Draft: `citemsp`: Citation Locators for LaTeX

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Abstract

I present `citemsp`, a LaTeX package extension of `biblatex` with per-key locators rendered as compact superscript/subscript pairs on numeric citation labels[1]. The default locators are section (§) and paragraph (¶), while an extensible prefix registry provides additional types – the user can register custom prefixes with `\citemspprefix`. The package requires no external dependencies beyond `biblatex`, `graphicx`, and `amssymb`.

1 Syntax and Usage

The `\citemsp` command embeds precise locators directly in multi-source citations. It accepts section (§) and paragraph (¶) locators by default, plus seven built-in prefix types – equation (eq.), box (□), definition (≜), figure (⊡), footnote (†), page (pp.), and table (⊞) – via a slash-delimited syntax with single-letter prefixes.

The command accepts a comma-separated list of citation keys with optional locators using forward-slash delimiters. By default, the first slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type:

`\citemsp{key}`	⟶	[4]
`\citemsp{key/section}`	⟶	[$4^{\S 6.2}$]
`\citemsp{key/section/paragraph}`	⟶	[$4^{\S 6.2}_{\P 3}$]
`\citemsp{key/section/e4.3}`	⟶	[$4^{\S 6.2}_{\text{eq. } 4.3}$]
`\citemsp{key/b21.1}`	⟶	[$3^{\square 21.1}$]
`\citemsp{key/section/b8.1}`	⟶	[$3^{\S 8.4}_{\square 8.1}$]

The seven built-in prefixes are listed in Table 1, and the user can register custom prefixes with `\citemspprefix{letter}{label}`.

2 Examples

[1] The package is available at https://github.com/apostolostsampodimos/citemsp and on CTAN.

Prefix	Locator	Render	Output
b	Box	□	[3$^{\square 21.1}$]
d	Definition	≜	[4$^{\S 3.2}_{\triangleq 5}$]
e	Equation	eq.	[4$^{\S 6.2}_{\text{eq. } 4.3}$]
f	Figure	⊡	[4$^{\S 6.1}_{\square 3}$]
n	Footnote	†	[2$^{\S 1}_{\dagger 7}$]
p	Page	pp.	[4$^{\text{pp. } 42}$]
t	Table	⊞	[3$^{\S 8.4}_{\boxplus 2}$]

Table 1: Registry: each single-letter prefix maps to a typographical symbol rendered before the locator number.

2.1 Single Source

For a single reference with precise location:

```
The Schwarzschild metric~\citemsp{wald1984/6.1/2} describes
spherically symmetric vacuum solutions.
```

Output: The Schwarzschild metric [4$^{\S 6.1}_{\P 2}$] describes spherically symmetric vacuum solutions.

2.2 Multiple Sources

For claims requiring verification across multiple independent sources:

```
Black hole thermodynamics~\citemsp{hawking1974/1/2, wald1984/12.5/1}
establishes a deep connection between gravity and quantum mechanics.
```

Output: Black hole thermodynamics [2$^{\S 1}_{\P 2}$, 4$^{\S 12.5}_{\P 1}$] establishes a deep connection between gravity and quantum mechanics.

2.3 Partial Locators

Not all entries require full locators:

```
General covariance~\citemsp{einstein1915, wald1984/4, hawking1974/1/2}
forms the foundation of general relativity.
```

Output: General covariance [1, 4$^{\S 4}$, 2$^{\S 1}_{\P 2}$] forms the foundation of general relativity.

2.4 Equation Locator

Use the `e` prefix to reference a specific equation:

```
The field equations~\citemsp{wald1984/3.2/e4.3} can be derived
from a variational principle.
```

Output: The field equations [4$^{\S 3.2}_{\text{eq. } 4.3}$] can be derived from a variational principle.

2.5 Box Locator

Use the `b` prefix for box environments, as used extensively in Misner, Thorne & Wheeler's *Gravitation* [3]:

```
The stress-energy tensor~\citemsp{mtw1973/b21.1} is discussed
in a self-contained box.
```

Output: The stress-energy tensor [3[□21.1]] is discussed in a self-contained box.

2.6 Definition Locator

Use the `d` prefix to reference a specific definition:

```
A globally hyperbolic spacetime~\citemsp{wald1984/8.1/d3}
satisfies strong causality with compact causal diamonds.
```

Output: A globally hyperbolic spacetime [4[§8.1]$_{\triangleq 3}$] satisfies strong causality with compact causal diamonds.

2.7 Figure Locator

Use the `f` prefix to point the reader to a specific figure:

```
The Penrose diagram~\citemsp{hawking1974/1/f2} illustrates
the causal structure near the event horizon.
```

Output: The Penrose diagram [2[§1]$_{\square 2}$] illustrates the causal structure near the event horizon.

2.8 Footnote Locator

Use the `n` prefix to cite a specific footnote in the source:

```
The sign convention is clarified in a
footnote~\citemsp{wald1984/3.4/n2}.
```

Output: The sign convention is clarified in a footnote [4[§3.4]$_{\dagger 2}$].

2.9 Page Locator

Use the `p` prefix to reference a specific page:

```
The ADM formalism~\citemsp{mtw1973/p520} decomposes
spacetime into spatial hypersurfaces.
```

Output: The ADM formalism [3[pp. 520]] decomposes spacetime into spatial hypersurfaces.

2.10 Table Locator

Use the `t` prefix to reference a specific table:

```
The Petrov classification~\citemsp{wald1984/7.3/t1} organises
the Weyl tensor into algebraic types.
```

Output: The Petrov classification [$4^{\S 7.3}_{\boxplus 1}$] organises the Weyl tensor into algebraic types.

2.11 Mixed Locator Types

All locator types can be freely mixed in a single call:

```
See~\citemsp{wald1984/6.1/2, mtw1973/b21.1, hawking1974/1/e4.3}
for further details.
```

Output: See [$4^{\S 6.1}_{\P 2}$, $3^{\square 21.1}$, $2^{\S 1}_{\text{eq. 4.3}}$] for further details.

2.12 Custom Prefixes

Users can register new prefix types with `\citemspprefix`. For example, a cosmology paper might define a prefix for "window function" references:

```
\citemspprefix{w}{w.}
The top-hat window~\citemsp{wald1984/4.2/w3} is
the simplest smoothing kernel.
```

Output: The top-hat window [$4^{\S 4.2}_{\text{w.3}}$] is the simplest smoothing kernel.

Similarly, a prefix for axioms can use a math symbol:

```
\citemspprefix{a}{\ensuremath{\vdash}}
The equivalence principle~\citemsp{mtw1973/16.4/a2}
underpins all metric theories of gravity.
```

Output: The equivalence principle [$3^{\S 16.4}_{\vdash 2}$] underpins all metric theories of gravity.

3 Relation to Biblatex

The standard `biblatex` provides locators via its postnote mechanism (e.g., `\autocite[§3.2]{wald1984}`), but `\citemsp` provides a compact way of guiding the reader to the precise location within the cited work, directly in the running text.

4 Implementation

1. **Locator scaling.** `\citemspscale` (default 0.35) and `\citesize` control the size of locator glyphs via `\scalebox` from `graphicx`.

2. **Citation rendering.** A custom `biblatex` cite command, `\citelinksp`, declared via `\DeclareCiteCommand`, prints the hyperlinked citation number and conditionally attaches superscript/subscript locators through the dispatcher. When `hyperref` draws bordered link boxes (`colorlinks=false`), a horizontal kern prevents overlap.

3. **Entry parsing.** The user-facing `\citemsp` uses `\forcsvlist` to iterate over comma-separated entries. Each entry is split at forward slashes using a delimited `\def` pattern, extracting the citation key and up to two locator slots before passing them to the renderer.

References

[1] Albert Einstein. “Die Feldgleichungen der Gravitation”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1915), pp. 844–847.

[2] Stephen W. Hawking. “Black hole explosions?” In: *Nature* 248 (1974), pp. 30–31.

[3] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman, 1973.

[4] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.